



# Saint Ignatius' College, Riverview

## Mathematics Assessment Task

### 2023

|                             |
|-----------------------------|
| Year 12                     |
| Mathematics Extension 1     |
| Task 4                      |
| Trial HSC                   |
| Date: Friday 18 August 2023 |

| General Instructions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Exam Sections:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>Reading Time: 10 minutes</b></li> <li>• <b>Time Allowed: 2 hours</b></li> <li>• Write using black pen</li> <li>• NESAs approved calculators may be used</li> <li>• Attempt all questions in the booklets provided.</li> <li>• Use a new Answer Booklet for each of questions 11, 12, 13 and 14.</li> <li>• Write <b>your name</b> and <b>your teacher's code</b> in the positions indicated on the Answer booklets</li> <li>• Marks may not be awarded for missing or carelessly arranged working.</li> </ul> <p><b>Teachers :</b></p> <ul style="list-style-type: none"> <li>• Mr R Maxwell                      <b>REM</b></li> <li>• Mr D Reidy                         <b>DPR</b></li> <li>• Mr N Mushan                      <b>NHM</b></li> <li>• Mr P Collins                        <b>PPC</b></li> </ul> | <p><b>Section A</b> <span style="float: right;"><b>10 marks</b></span></p> <p>Multiple Choice</p> <p><b>Section B</b></p> <p>Short Answer</p> <p><b>Question 11</b> <span style="float: right;"><b>15 marks</b></span></p> <p><b>Question 12</b> <span style="float: right;"><b>15 marks</b></span></p> <p><b>Question 13</b> <span style="float: right;"><b>15 marks</b></span></p> <p><b>Question 14</b> <span style="float: right;"><b>15 marks</b></span></p> <hr/> <p><b>Total</b> <span style="float: right;"><b>70 Marks</b></span></p> |

## SECTION A: Multiple Choice

Answer Questions 1 to 10 on the Multiple Choice Answer Grid

1. What is the remainder when  $P(x) = -2x^3 + 3x^2 - 6x + 4$  is divided by  $x + 2$ .

- A. 44
- B. -12
- C. 12
- D. 20

2. What is the derivative of  $\tan^{-1}(4x-1)$ ?

- A.  $\frac{1}{16x^2 - 8x + 2}$
- B.  $\frac{16x - 4}{8x^2 - 4x + 1}$
- C.  $\frac{4}{8x^2 - 4x + 1}$
- D.  $\frac{2}{8x^2 - 4x + 1}$

3. The domain for  $f(x) = 3\cos^{-1} 4x$  is:

- A.  $\left[\frac{-3}{4}, \frac{3}{4}\right]$
- B.  $\left[\frac{-1}{3}, \frac{1}{3}\right]$
- C.  $\left[\frac{-1}{4}, \frac{1}{4}\right]$
- D.  $[-1, 1]$

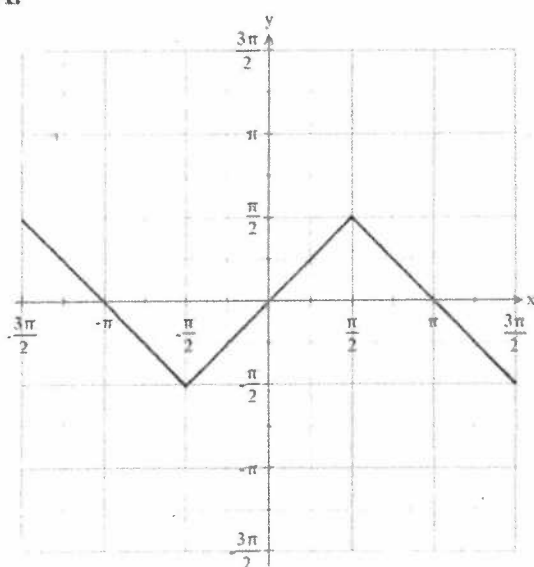
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. | <p>What is the value of <math>\sin 2x</math> given that <math>\sin x = \frac{\sqrt{3}}{2}</math> and <math>x</math> is obtuse?</p> <p>A. <math>-\frac{\sqrt{3}}{4}</math></p> <p>B. <math>-\frac{\sqrt{3}}{2}</math></p> <p>C. <math>\frac{\sqrt{3}}{4}</math></p> <p>D. <math>\frac{\sqrt{3}}{2}</math></p>                                                                                                                                                                                  |
| 5. | <p>A curve <math>C</math> has parametric equations <math>x = \cos^2 t</math> and <math>y = 2 \sin^2 t</math> for <math>-\infty &lt; t &lt; \infty</math>.</p> <p>What is the Cartesian equation and domain of <math>C</math>?</p> <p>A. <math>y = 1 - x</math> for <math>[0, 1]</math></p> <p>B. <math>y = 2 - 2x</math> for <math>(-\infty, \infty)</math></p> <p>C. <math>y = 2 - 2x</math> for <math>[0, 1]</math></p> <p>D. <math>y = 1 - x</math> for <math>(-\infty, \infty)</math></p> |
| 6. | <p>Which of the following integrals is equivalent to <math>\int \cos^2 4x \, dx</math>?</p> <p>A. <math>\frac{1}{2} \int (x + \frac{1}{8} \sin 8x) \, dx</math></p> <p>B. <math>\frac{1}{2} \int (x - \frac{1}{8} \sin 8x) \, dx</math></p> <p>C. <math>\frac{1}{2} \int (1 + \cos 8x) \, dx</math></p> <p>D. <math>\frac{1}{2} \int (1 - \cos 8x) \, dx</math></p>                                                                                                                           |

7. For the two vectors  $\vec{AB}$  and  $\vec{AC}$  it is known that  $\vec{AB} \cdot \vec{AC} > 0$ . Which of the following statements must be true?

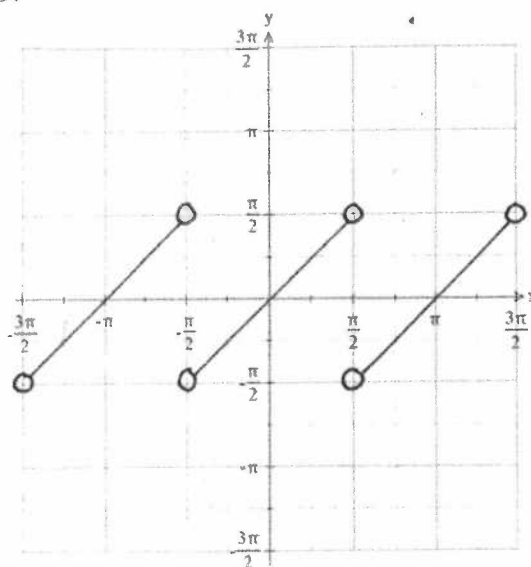
- A. Either  $\vec{AB}$  is positive and  $\vec{AC}$  is positive, or,  $\vec{AB}$  is negative and  $\vec{AC}$  is negative.
- B. The angle between  $\vec{AB}$  and  $\vec{AC}$  is acute.
- C. The product  $|\vec{AB}| |\vec{AC}|$  is negative.
- D. The points  $A$ ,  $B$  and  $C$  are collinear.

8. Which graph represents the function  $y = \tan^{-1}(\tan x)$ ?

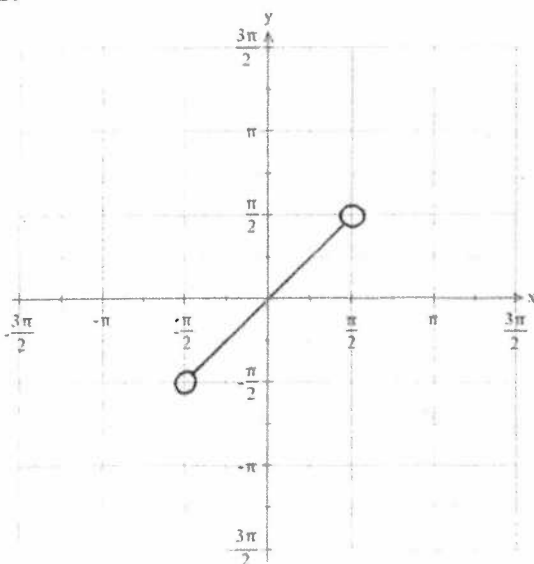
A.



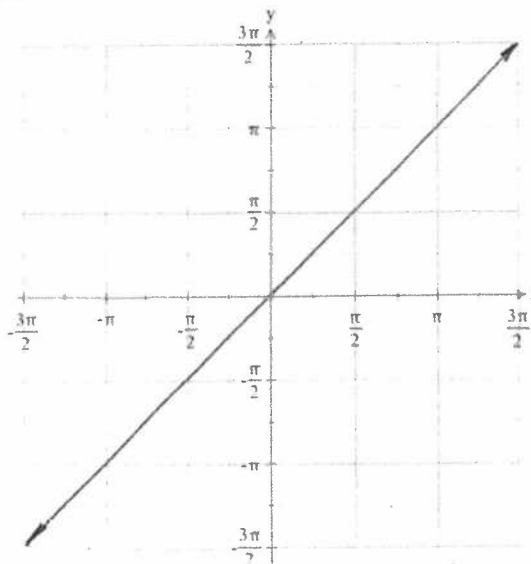
B.



C.



D.

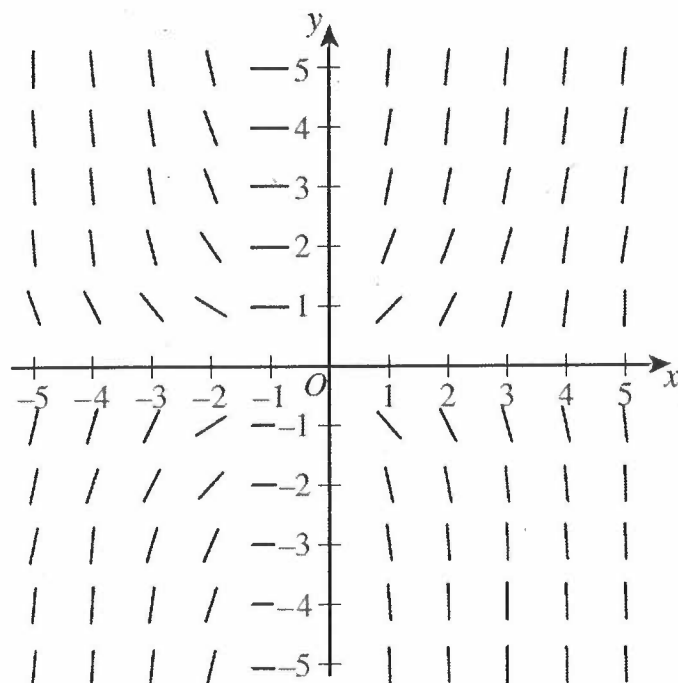


9. Let  $P(x) = x^2 + bx + c$  where  $b$  and  $c$  are constants. The zeros of  $P(x)$  are  $\alpha$  and  $\alpha + 1$ .

What are the correct expressions for  $b$  and  $c$  in terms of  $\alpha$ ?

- A.  $b = -(2\alpha + 1)$  and  $c = \alpha^2 + \alpha$
- B.  $b = 2\alpha + 1$  and  $c = \alpha^2 + \alpha$
- C.  $b = \alpha^2 + \alpha$  and  $c = -(2\alpha + 1)$
- D.  $b = \alpha^2 + \alpha$  and  $c = 2\alpha + 1$

10. The direction (slope) field for a first order differential equation is shown.



Which of the following could be the differential equation represented?

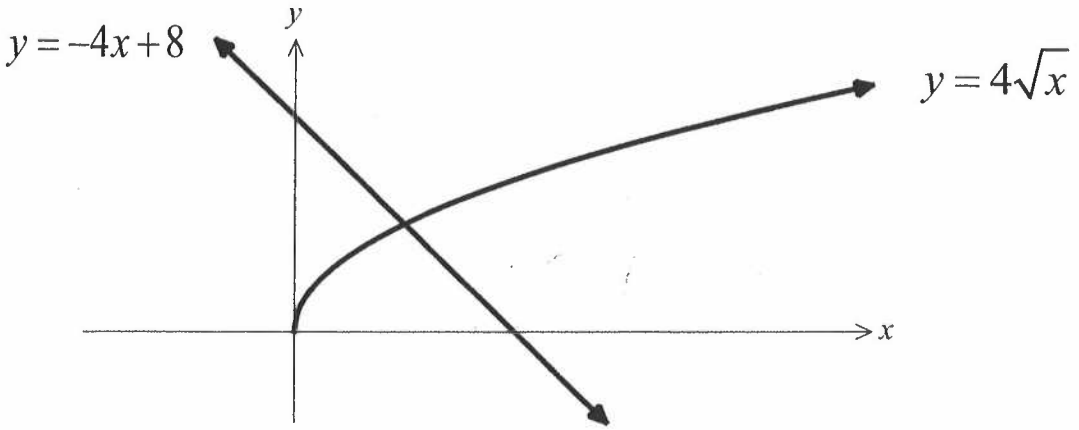
- A.  $\frac{dy}{dx} = (x+1)^3$
- B.  $\frac{dy}{dx} = x(y+1)$
- C.  $\frac{dy}{dx} = (x-1)y$
- D.  $\frac{dy}{dx} = (x+1)y$

QUESTION 11

Use a new writing booklet

(15 marks)

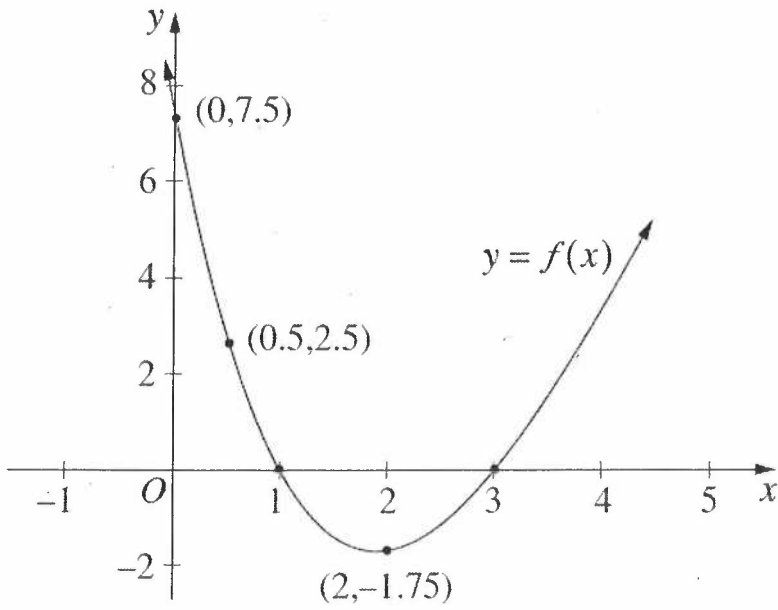
|    |                                                                                                                                                                                                                                                                 | Marks               |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| a. | Find the anti-derivative of $\frac{1}{\sqrt{4-9x^2}}$                                                                                                                                                                                                           | 2                   |
| b. | i. Find the exact value of $\sin \frac{23\pi}{12}$ .<br>(Express answer as a single fraction with a rational denominator)                                                                                                                                       | 2                   |
|    | ii. Find the exact value of $\sin^{-1}(\sin \frac{23\pi}{12})$                                                                                                                                                                                                  | 1                   |
| c. | Consider the vectors $\underline{u} = \underline{i} - 2\underline{j}$ and $\underline{v} = a\underline{i} + 2\underline{j}$<br>i. Find the value of $a$ if $\underline{u}$ and $\underline{v}$ are perpendicular.                                               | 1                   |
|    | ii. Find the values of $a$ if the angle between $\underline{u}$ and $\underline{v}$ is $60^\circ$ .<br>Give answers correct to 1 decimal place                                                                                                                  | 3                   |
| d. | Given $\underline{u} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 6 \\ -8 \end{pmatrix}$<br>i. Find $\text{proj}_{\underline{u}} \underline{v}$                                                                                 | 2                   |
|    | ii. Find the direction of $\text{proj}_{\underline{u}} \underline{v}$<br>Give answer to nearest degree.                                                                                                                                                         | 1                   |
| e. | Consider the cubic equation $6x^3 - 9x^2 + 4x - 3 = 0$ with roots $\alpha$ , $\beta$ and $\gamma$ .<br>Find the value of:<br><br>i. $\alpha + \beta + \gamma$<br><br>ii. $\alpha\beta + \alpha\gamma + \beta\gamma$<br><br>iii. $\alpha^2 + \beta^2 + \gamma^2$ | 1<br><br>1<br><br>1 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| a. | <p>The area enclosed by the curve <math>y = 4\sqrt{x}</math>, the line <math>y = -4x + 8</math> and the <math>x</math> axis is rotated about the <math>x</math> axis.</p>  <p>i. Find the coordinates of the point where <math>y = 4\sqrt{x}</math> and <math>y = -4x + 8</math> intersect.</p> <p>ii. Find the exact volume of the solid of revolution.</p> | <p>1</p> <p>3</p> |
| b. | <p>Solve: <math>\frac{3x}{x+3} \leq -2</math></p>                                                                                                                                                                                                                                                                                                                                                                                              | <p>3</p>          |
| c. | <p>Use the substitution <math>u = 1 + 2 \tan x</math> to evaluate <math>\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx</math></p>                                                                                                                                                                                                                                                                                               | <p>3</p>          |
| d. | <p>i. Express <math>\cos x - \sqrt{3} \sin x</math> in the form <math>R \cos(x + \alpha)</math></p> <p>ii. Hence, state the greatest value of the function: <math>f(x) = \cos x - \sqrt{3} \sin x + 10</math></p>                                                                                                                                                                                                                              | <p>1</p> <p>1</p> |
| e. | <p>Consider the function <math>y = 2x(x+6)^6</math>. Find the derivative of the inverse function.</p> <p>Give answer in terms of <math>y</math>.</p>                                                                                                                                                                                                                                                                                           | <p>3</p>          |

# QUESTION 13

Use a new writing booklet

(15 marks)

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks               |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| a. | <p>The diagram below is a sketch of the graph <math>y = f(x)</math>.</p>  <p>i. Sketch the graph of <math>y = f( x )</math> showing all important features, together with the location of the points corresponding to the labelled points on the original sketch.</p> <p>ii. Sketch the graph of <math>y = \frac{1}{f(x)}</math> showing all important features, together with the location of the points corresponding to the labelled points on the original sketch.</p>     | 2<br><br>2          |
| b. | Solve the differential equation $f'(x) = 4x^3 y^2$ given $f(2) = 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                   |
| c. | Use mathematical induction to prove $7^{2n} + 7^n + 4$ is divisible by 6 for all integers $n \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                   |
| d. | <p>A metal is heated to <math>200^\circ\text{C}</math> and then left to cool, which it does according to the formula <math>\frac{dT}{dt} = -0.05(T - 23)</math> where <math>T</math> is the temperature in <math>^\circ\text{C}</math> and <math>t</math> is time in minutes.</p> <p>i. Solve the differential equation. (Make <math>T</math> the subject)</p> <p>ii. Find the temperature after 20 minutes. (Answer to 2 decimal places)</p> <p>iii. Find the time when the temperature reaches <math>40^\circ\text{C}</math>. (Answer to 2 decimal places)</p> | 3<br><br>1<br><br>1 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks                                              |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| a. | <p>A particle is projected from a point <math>O</math> on level ground with a speed of <math>25 \text{ ms}^{-1}</math> at an angle <math>\theta</math> to the horizontal. At time <math>T</math> seconds, the particle passes through the point <math>B(15, 2)</math>.</p> <p>The displacement vector <math>R(t)</math> for the motion of the particle is:</p> $R(t) = (Vt \cos \theta) \mathbf{i} + \left(-\frac{1}{2}gt^2 + Vt \sin \theta\right) \mathbf{j} \quad (\text{Do Not Prove this!})$ <p>Where <math>t</math> is the time in seconds after projection, <math>\mathbf{i}</math> and <math>\mathbf{j}</math> are unit vectors of magnitude 1 metre and <math>g</math> is the acceleration due to gravity <math>= 10 \text{ ms}^{-2}</math>.</p> <p>i. By considering the horizontal component of the particle's motion, show that <math>T = \frac{3}{5} \sec \theta</math>.</p> <p>ii. By considering the vertical component of the particle's motion and, using the result from part i, show that <math>9 \tan^2 \theta - 75 \tan \theta + 19 = 0</math>.</p> <p>iii. Find the particle's least possible flight time from <math>O</math> to <math>B</math>. Give your answer correct to 2 decimal places.</p> | <p></p> <p></p> <p></p> <p>1</p> <p>2</p> <p>2</p> |
| b. | <p>Two unbiased dice are rolled.</p> <p>i. Find the probability that the difference of the numbers on the dice is 1. (answer to 2 decimal places)</p> <p>ii. The two dice are rolled 60 times. For the binomial distribution of rolling a difference of 1, find the mean and the variance. (answer to 2 decimal places)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p></p> <p>1</p> <p>2</p>                          |
| c. | <p>i. If <math>t = \tan \frac{A}{2}</math>, express <math>3 \sin A - 4 \cos A - 4</math> in terms of <math>t</math></p> <p>ii. Hence solve <math>3 \sin A - 4 \cos A = 4</math> for <math>0 \leq A \leq 2\pi</math> (answer to 2 decimal places)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>2</p> <p>2</p>                                  |
| d. | <p>Given <math>\sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} = \frac{1 - \tan x}{1 + \tan x}</math></p> <p>Show that <math>\tan \frac{\pi}{8} = \sqrt{2} - 1</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p></p> <p>3</p>                                   |

END OF TASK



Saint Ignatius' College, Riverview  
Mathematics Assessment Task  
2023

|                             |
|-----------------------------|
| Year 12                     |
| Mathematics Extension 1     |
| Task 4                      |
| Trial HSC                   |
| Date: Friday 18 August 2023 |

# Suggested Solutions

| Exam Sections:                 |                 | Marker |
|--------------------------------|-----------------|--------|
| <b>Section A</b>               | 10 marks        |        |
| Multiple Choice                |                 | JPN    |
| <b>Section B: Short Answer</b> |                 |        |
| Question 11                    | 15 marks        | PPC    |
| Question 12                    | 15 marks        | NHM    |
| Question 13                    | 15 marks        | DPR    |
| Question 14                    | 15 marks        | REM    |
| <b>Total</b>                   | <b>70 marks</b> |        |

## Multiple Choice

1. A
2. D
3. C
4. B
5. C
6. C
7. B
8. B
9. A
10. D

## Multiple Choice Solutions

1.  $P(-2) = -2(-2)^3 + 3(-2)^2 - 6(-2) + 4$   
 $= 44$

ANSWER (A)

2. from formula sheet

$$\tan^{-1}(4x-1) = \frac{4}{(1-4x)^2 + 1}$$

$$= \frac{4}{16x^2 - 8x + 1 + 1}$$

$$= \frac{4}{16x^2 - 8x + 2}$$

$$= \frac{2}{8x^2 - 4x + 1}$$

ANSWER (D)

3.  $f(x) = 3 \cos^{-1} 4x$

Domain is affected by the 4

$\therefore$  Domain  $[-\frac{1}{4}, \frac{1}{4}]$  ANSWER (C)

4.  $\sin x = \frac{\sqrt{3}}{2}$   $x$  is obtuse  $\therefore \cos x = -\frac{1}{2}$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{\sqrt{3}}{2} \times -\frac{1}{2}$$

$$= -\frac{\sqrt{3}}{2}$$

ANSWER (B)

5.  $2x + y = 2 \cos^2 t + 2 \sin^2 t$   
 $= 2(\cos^2 t + \sin^2 t)$

$$y = 2 - 2x$$

$$\text{Since } x = \cos^2 t \quad 0 \leq x \leq 1$$

$$\therefore y = 2 - 2x \quad [0, 1] \quad \text{ANSWER (C)}$$

6. from reference sheet

$$\cos^2 4x = \frac{1}{2}(1 + \cos 8x)$$

$$\begin{aligned}\therefore \int \cos^2 4x &= \int \frac{1}{2}(1 + \cos 8x) dx \\ &= \frac{1}{2} \int (1 + \cos 8x) dx \quad \text{ANSWER (C)}\end{aligned}$$

7.  $\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$

since  $\vec{AB} \cdot \vec{AC} > 0$  and  $|\vec{AB}| \times |\vec{AC}| > 0$

$$\therefore \cos \theta > 0$$

$$\therefore \theta \text{ must be acute.} \quad \text{ANSWER (B)}$$

8.  $\tan^{-1}(\tan x)$  is undefined when  $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \text{ etc}$   
but defined for all other values  
ANSWER (B)

9.  $\alpha + \beta = -\frac{b}{a}$

$$\alpha\beta = \frac{c}{a}$$

$$\alpha + \alpha + 1 = -b$$

$$\alpha(\alpha + 1) = c$$

$$2\alpha + 1 = -b$$

$$c = \alpha^2 + \alpha$$

$$\therefore b = -(2\alpha + 1)$$

$$\text{ANSWER (A)}$$

10. ANSWER (D)

Test numerous points and eliminate

eg.  $x = -5$   $y = -1$  gradient positive  $\therefore$  not A

$x = -3$   $y = -1$  gradient zero  $\therefore$  not B

$x = 1$   $y = 1$  gradient zero  $\therefore$  not C

Answer (D)

# QUESTION 11

$$\begin{aligned} \textcircled{a} \int \frac{1}{\sqrt{4-9x^2}} dx &= \int \frac{1}{\sqrt{9(\frac{4}{9}-x^2)}} dx \\ &= \int \frac{1}{3\sqrt{\frac{4}{9}-x^2}} dx \\ &= \frac{1}{3} \int \frac{1}{\sqrt{(\frac{2}{3})^2-x^2}} dx \\ &= \boxed{\frac{1}{3} \sin^{-1} \frac{3x}{2} + C} \end{aligned}$$

FROM REF. SHEET.

$$\int \frac{f'(x)}{\sqrt{a^2-(f(x))^2}} dx = \sin^{-1} \frac{f(x)}{a} + C$$

$$\therefore \frac{1}{3} \int \frac{3 \times 1}{\sqrt{2^2-(3x)^2}} = \frac{1}{3} \sin^{-1} \frac{3x}{2} + C$$

Marker's Comments

Many missed the  $\frac{1}{3}$  out front or put 3 out front!

$$\begin{aligned} \textcircled{b} \textcircled{i} \sin \frac{23\pi}{12} &= \sin \left( \frac{5\pi}{3} + \frac{\pi}{4} \right) \\ &= \sin \frac{5\pi}{3} \cos \frac{\pi}{4} + \cos \frac{5\pi}{3} \sin \frac{\pi}{4} \\ &= -\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{1-\sqrt{3}}{2\sqrt{2}} \\ &= \boxed{\frac{\sqrt{2}-\sqrt{6}}{4}} \end{aligned}$$

OR correct similar

$$\begin{aligned} \textcircled{b} \textcircled{ii} \sin^{-1} \left( \sin \frac{23\pi}{12} \right) &= \sin^{-1} \left( \sin -\frac{\pi}{12} \right) \\ &= \boxed{-\frac{\pi}{12}} \end{aligned}$$

Range of  $y = \sin^{-1}$   
 $-\frac{\pi}{2} \leq y < \frac{\pi}{2}$

$\therefore$  Not defined for  $\frac{23\pi}{12}$  but is for  $-\frac{\pi}{12}$ .

$$\begin{aligned} \textcircled{c} \textcircled{i} \underline{u} \cdot \underline{v} &= x_1x_2 + y_1y_2 \\ \underline{u} \cdot \underline{v} &= 1 \times a - 2 \times 2 \\ &= a - 4 \end{aligned}$$

perpendicular  $\therefore a - 4 = 0$

$$\therefore \boxed{a = 4}$$

$$\begin{aligned} \textcircled{ii} |\underline{u}| &= \sqrt{1+(-2)^2} = \sqrt{5} \\ |\underline{v}| &= \sqrt{a^2+2^2} = \sqrt{a^2+4} \end{aligned}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$a - 4 = \sqrt{5} \times \sqrt{a^2+4} \times \cos 60^\circ$$

$$2a - 8 = \sqrt{5a^2+20}$$

$$4a^2 - 32a + 64 = 5a^2 + 20$$

$$0 = a^2 + 32a - 44$$

$$a = \frac{-32 \pm \sqrt{32^2 - 4(1)(-44)}}{2 \times 1}$$

$$\boxed{a = 1.3 \text{ or } -33.3}$$

The New Fx 8200 will do it for you

(i) Very well done

Many problems with algebraic manipulations.

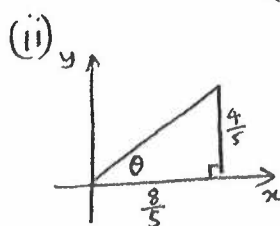
# QUESTION 11

## Marker's Comments

$$\begin{aligned} \textcircled{d} \text{ (i)} \quad \underline{u} \cdot \underline{v} &= 4 \times 6 + 2 \times -8 \\ &= 24 - 16 \\ &= 8 \end{aligned}$$

$$\begin{aligned} |\underline{u}| &= \sqrt{4^2 + 2^2} \\ &= \sqrt{20} \end{aligned}$$

$$\begin{aligned} \text{proj}_{\underline{u}} \underline{v} &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|^2} \underline{u} \\ &= \frac{8}{20} \times \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{8}{5} \\ \frac{4}{5} \end{pmatrix} \end{aligned}$$



$$\tan \theta = \frac{4}{5} \div \frac{8}{5}$$

$$\tan \theta = \frac{1}{2}$$

$$\theta = 26.565^\circ$$

$$\text{Direction} = \theta = \boxed{27^\circ}$$

also accept  $\frac{2}{5} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

This formula is not on the Reference sheet. You need to learn it, and apply it, accurately.

Many who were in-correct in part(i) still managed part(ii) as the projection and  $\underline{u}$  have the same angle!

$$\textcircled{e} \quad 6x^3 - 9x^2 + 4x - 3 = 0$$

$$\text{(i)} \quad \alpha + \beta + \gamma = -\frac{b}{a} = \frac{9}{6} = \boxed{\frac{3}{2}}$$

$$\text{(ii)} \quad \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{4}{6} = \boxed{\frac{2}{3}}$$

$$\begin{aligned} \text{(iii)} \quad (\alpha + \beta + \gamma)^2 &= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma + \beta\gamma + \gamma^2 \\ &= \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \end{aligned}$$

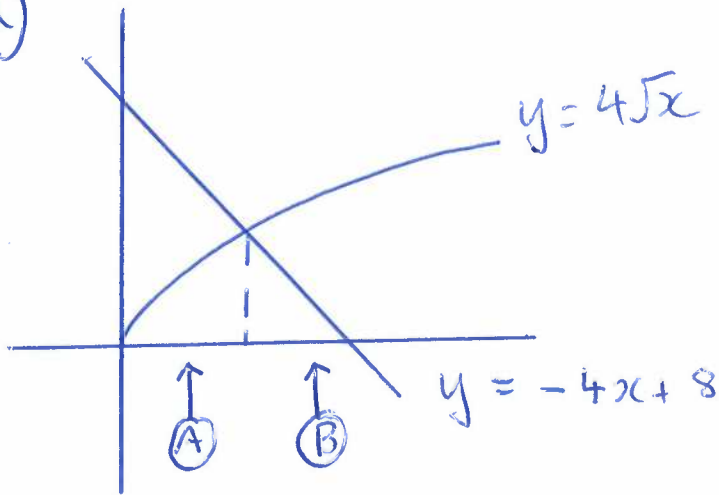
$$\begin{aligned} \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(\frac{3}{2}\right)^2 - 2\left(\frac{2}{3}\right) \\ &= \frac{9}{4} - \frac{4}{3} \\ &= \boxed{\frac{11}{12}} \end{aligned}$$

✓ Page 1 of Reference sheet!  
✓ (i)(ii) Generally well done

(iii) No so well done

## QUESTION 12

a)



$$\begin{aligned}
 \text{i) } y &= 4\sqrt{x} & 4\sqrt{x} &= -4x + 8 \\
 y &= -4x + 8 & 16x &= 16x^2 - 64x + 64 \\
 & & 0 &= 16x^2 - 80x + 64 \\
 & & &= x^2 - 5x + 4 \\
 & & &= (x-1)(x-4)
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= 1 & x &= 4 \\
 y &= 4 & y &= -8
 \end{aligned}$$

$\therefore$  Point of Intersection

$$(1, 4) \leftarrow \text{1mk}$$

$$\text{ii) } V = \pi \int_a^b y^2 dx$$

$$\begin{aligned}
 \text{Volume (A) } V &= \pi \int_0^1 16x dx \\
 &= \pi [8x^2]_0^1 \\
 &= 8\pi \text{ units}^3 \leftarrow \text{1mk}
 \end{aligned}$$

Fairly well done.

Some careless errors.

Volume (B)

$$V = \pi \int_1^2 (-4x + 8)^2 dx$$

$$= \pi \int_1^2 (16x^2 - 64x + 64) dx$$

$$= \pi \left[ \frac{16x^3}{3} - 32x^2 + 64x \right]_1^2$$

$$= \pi \left[ \left( \frac{128}{3} \right) - \left( \frac{112}{3} \right) \right]$$

$$= \frac{16\pi}{3} \text{ Units}^3 \quad \leftarrow \text{1mk}$$

$$\text{Volume Total} = 8\pi + \frac{16\pi}{3}$$

$$= \frac{40\pi}{3} \text{ u}^3 \quad \leftarrow \text{1mk}$$

(4)

b)  $\frac{3x}{x+3} \leq -2$

$$3x(x+3) \leq -2(x+3)^2$$

$$2(x+3)^2 + 3x(x+3) \leq 0$$

$$(x+3)(2(x+3) + 3x) \leq 0$$

$$(x+3)(5x+6) \leq 0$$



$$-3 < x \leq -\frac{6}{5}$$

Many need more time revising the volumes topic.

Too many had forgotten method.

$$-3 < x \leq -\frac{6}{5}$$

↑  
1mk

↑  
1mk

↑  
1mk

(3)

c)

$$\int_0^{\frac{\pi}{4}} \frac{1}{(1+2\tan x)^2 \cos^2 x} dx$$

$$u = 1 + 2\tan x$$

$$\frac{du}{dx} = 2\sec^2 x$$

$$du = 2\sec^2 x dx$$

$$\frac{1}{2} du = \frac{1}{\cos^2 x} dx$$

$$x=0 \quad u=1$$

$$x=\frac{\pi}{4} \quad u=3$$

poorly done

$$\therefore \int_0^{\frac{\pi}{4}} \frac{1}{(1+2\tan x)^2 \cos^2 x} dx = \frac{1}{2} \int_1^3 \frac{1}{u^2} du$$

$$= \frac{1}{2} \int_1^3 u^{-2} du$$

$$= \frac{1}{2} \left[ -\frac{1}{u} \right]_1^3$$

$$= \frac{1}{2} \left[ \left( -\frac{1}{3} \right) - (-1) \right]$$

$$= \frac{1}{3}$$

mk for correct integral

mk.

3

$$d) \quad 1) \cos x - \sqrt{3} \sin x \\ = R \cos(x + \alpha)$$



$$R^2 = 1^2 + (\sqrt{3})^2 \quad \tan \alpha = \frac{\sqrt{3}}{1} \\ R = 2 \quad \alpha = \frac{\pi}{3}$$

Well done

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right) \leftarrow \text{mk}$$

$$1) \quad f(x) = \cos x - \sqrt{3} \sin x + 10 \\ = \underbrace{2 \cos\left(x + \frac{\pi}{3}\right)}_{\text{max} = 2} + 10$$

Well done

$$\therefore \text{Max} = 12 \quad \leftarrow \text{mk}$$

2

e)

$$y = 2x(x+6)^6$$

$$x = 2y(y+6)^6 \leftarrow \text{Imk}$$

$$\begin{aligned} \frac{dx}{dy} &= 2y[6(y+6)^5] + [y+6]^6 \cdot 2 \leftarrow \text{Imk for} \\ &= (y+6)^5 \cdot (12y + 2(y+6)) \end{aligned}$$

Correct use of product rule

$$= (y+6)^5 \cdot (14y + 12)$$

$$= 2(y+6)^5(7y+6)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2(y+6)^5(7y+6)} \leftarrow \text{Imk}$$

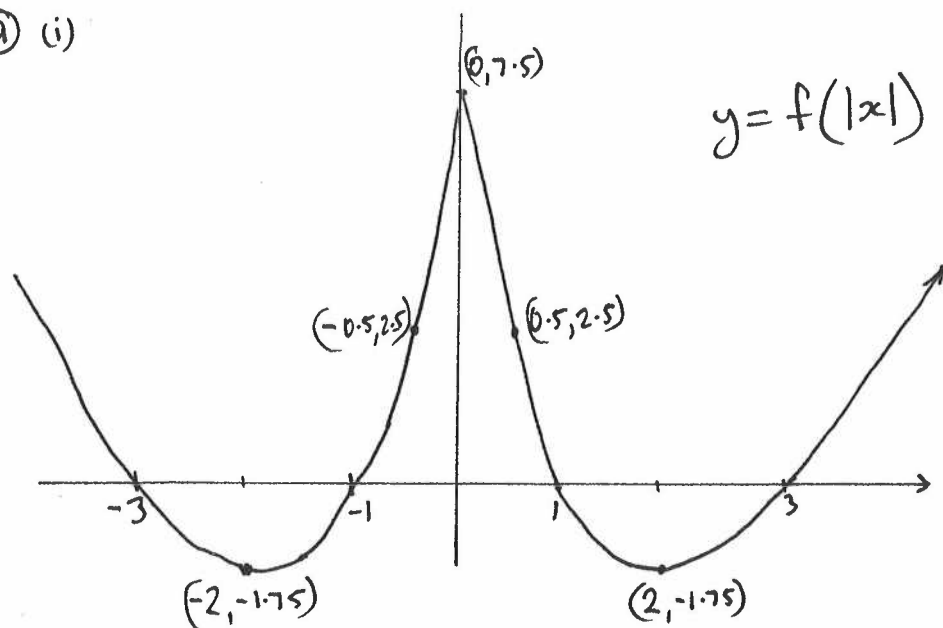
Desirable to go to  $\frac{dy}{dx} =$

but no penalty if left at  
CORRECT  $\frac{dx}{dy}$

# QUESTION 13

## Marker's Comments

(a) (i)

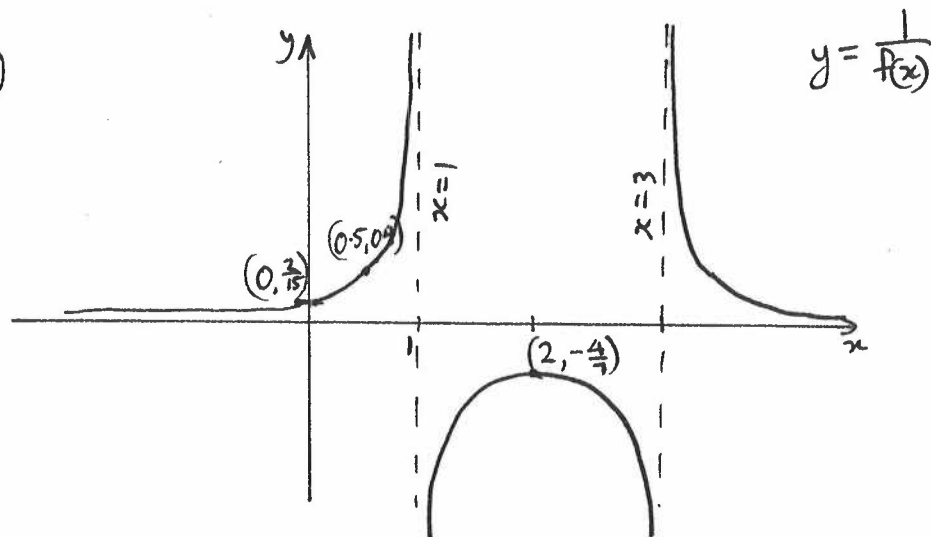


\* Well answered.

\* 1 mark for correct shape.

\* 1 mark for including coordinates of intercepts and turning points.

(ii)



\* Well answered

\* 1 mark for correct shape

\* 1 mark for including coordinates of intercepts and turning points.

$$(b) \frac{dy}{dx} = 4x^3 y^2$$

$$\frac{1}{y^2} dy = 4x^3 dx$$

$$\int y^{-2} dy = \int 4x^3 dx$$

$$\frac{y^{-1}}{-1} + c_1 = x^4 + c_2$$

$$-\frac{1}{y} = x^4 + k$$

$$\text{when } x=2, y=1$$

$$-\frac{1}{1} = 2^4 + k$$

$$-1 = 16 + k$$

$$k = -17$$

$$\therefore -\frac{1}{y} = x^4 - 17$$

$$y = \frac{1}{17 - x^4}$$

\* Well answered.

\* some students did not calculate the constant correctly but received carry on marks.

\* some students did not set up the differential correctly at the start but received carry on marks afterwards.

# QUESTION 13

Marker's Comments

① Prove  $7^{2n} + 7^n + 4$  is divisible by 6 for all integers  $n \geq 0$

Step 1: Prove true for  $n=0$

$$7^{2 \times 0} + 7^0 + 4 = 1 + 1 + 4 = 6 \text{ which is divisible by 6}$$

$$\therefore \text{true for } n=0$$

Step 2: Assume true for  $n=k$

$$\text{ie } 7^{2k} + 7^k + 4 = 6M \text{ where } M \text{ is a positive integer}$$

$$\text{ie } 7^{2k} = 6M - 7^k - 4$$

Step 3: Prove true for  $n=k+1$

$$\text{ie } 7^{2(k+1)} + 7^{(k+1)} + 4 = 6P \text{ where } P \text{ is a positive integer}$$

$$\text{LHS} = 7^{2k+2} + 7^{k+1} + 4$$

$$= 7^{2k} \times 7^2 + 7^{k+1} + 4$$

$$= 49(6M - 7^k - 4) + 7 \times 7^k + 4 \dots \text{from assumption in step 2}$$

$$= 294M - 49 \times 7^k - 196 + 7 \times 7^k + 4$$

$$= 294M - 42 \times 7^k - 192$$

$$= 6(49M - 7 \times 7^k - 32)$$

$$= 6P \text{ since } M \text{ and } k \text{ are integers.}$$

If true for  $n=k$ , then true for  $n=k+1$

Since it is true for  $n=0$ , then it is true for  $n=1$ .

If true for  $n=1$ , then it is true for  $n=2$  and so on

for all positive integers  $n \geq 0$

\* Well answered.

\* Some students

substituted  $n=1$

for the first step

and got penalised.

# QUESTION 13

## Marker's Comments

$$(d) (i) \frac{dT}{dt} = -0.05(T-23)$$

$$\frac{dt}{dT} = \frac{-1}{0.05(T-23)}$$

$$= \frac{-20}{T-23}$$

$$t = -20 \int \frac{1}{T-23} dT$$

$$= -20 \ln|T-23| + c$$

$$\text{when } t=0 \quad T=200$$

$$0 = -20 \ln|200-23| + c$$

$$c = 20 \ln 177$$

$$\therefore t = -20 \ln|T-23| + 20 \ln 177$$

$$t = 20(\ln 177 - \ln|T-23|)$$

$$t = 20 \ln \frac{177}{T-23}$$

$$\frac{t}{20} = \ln \frac{177}{T-23}$$

$$e^{\frac{t}{20}} = e^{\ln \frac{177}{T-23}}$$

$$\frac{177}{T-23} = e^{\frac{t}{20}}$$

$$177 = (T-23)e^{\frac{t}{20}}$$

$$177e^{-\frac{t}{20}} = T-23$$

$$T = 23 + 177e^{-\frac{t}{20}}$$

$$T = 23 + 177e^{-0.05t}$$

$$(ii) \text{ when } t=20$$

$$T = 23 + 177e^{-0.05 \times 20}$$

$$T = 88.11^\circ \text{C}$$

$$(iii) \text{ when } T=40$$

$$40 = 23 + 177e^{-0.05t}$$

$$17 = 177e^{-0.05t}$$

$$\frac{17}{177} = e^{-0.05t}$$

$$\ln \frac{17}{177} = -0.05t$$

$$t = \frac{\ln \frac{17}{177}}{-0.05} = 46.86 \text{ minutes}$$

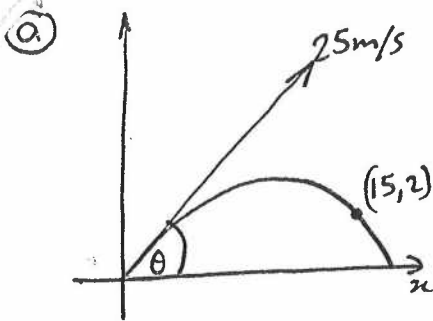
46 minutes 52 seconds

\* Well answered

\* Some students incorporated the ~~power~~ constant of integration into the exponential power when calculating and lost marks.

\* Well answered

\* Well answered.



When  $t = T$   
 $x = 15$

(i)  $x = Vt \cos \theta$   
 $15 = 25t \cos \theta$   
 $T = \frac{15}{25 \cos \theta}$  ✓

$T = \frac{3}{5} \sec \theta$  (as required)

(ii)  $y = Vt \sin \theta - \frac{1}{2} \times 10 \times t^2$

When  $t = T$   
 $y = 2$

$2 = 25 \times \frac{3}{5} \sec \theta \sin \theta - 5 \left( \frac{3}{5} \sec \theta \right)^2$

$2 = 15 \tan \theta - 5 \times \frac{9}{25} \sec^2 \theta$  → ✓ 1 Mark

$2 = 15 \tan \theta - \frac{9}{5} \sec^2 \theta$

$10 = 75 \tan \theta - 9(1 + \tan^2 \theta)$  → ✓ 1 Mark

$10 = 75 \tan \theta - 9 - 9 \tan^2 \theta$

$9 \tan^2 \theta - 75 \tan \theta + 19 = 0$   
 (as required)

(iii)  $\tan \theta = \frac{75 \pm \sqrt{75^2 - 4(9)(19)}}{2 \times 9}$   
 $= \frac{75 \pm \sqrt{4941}}{18}$

$\tan \theta = 0.2615418...$  or  $\tan \theta = 8.07179...$  → 1 Mark  
 $\theta = 14.6569...^\circ$  or  $\theta = 82.9377...^\circ$

$\therefore T = \frac{3}{5 \cos 14.656...^\circ}$   
 $= 0.62 \text{ seconds}$   
 (2d.p)

$T = \frac{3}{5 \cos 82.9377...^\circ}$   
 $= 4.88 \text{ seconds}$

$\therefore$  Least possible flight time from O to B

is 0.62 seconds ✓

Correct Answer.

# QUESTION 14

Marker's Comments

(b)

| (i) 2nd Die | 1st Die |   | Difference |   |   |   |  |  |
|-------------|---------|---|------------|---|---|---|--|--|
|             | 1       | 2 | 3          | 4 | 5 | 6 |  |  |
| 1           | 0       | 1 | 2          | 3 | 4 | 5 |  |  |
| 2           | 1       | 0 | 1          | 2 | 3 | 4 |  |  |
| 3           | 2       | 1 | 0          | 1 | 2 | 3 |  |  |
| 4           | 3       | 2 | 1          | 0 | 1 | 2 |  |  |
| 5           | 4       | 3 | 2          | 1 | 0 | 1 |  |  |
| 6           | 5       | 4 | 3          | 2 | 1 | 0 |  |  |

$$P(\text{difference} = 1) = \frac{10}{36} = \boxed{\frac{5}{18}} \text{ OR } 0.28 \rightarrow 1 \text{ Mark}$$

$$(ii) \text{ mean} = E(X) = 60 \times \frac{5}{18} = \frac{50}{3} = 16\frac{2}{3} = \boxed{16.6} \text{ OR } 60 \times 0.28 \rightarrow \checkmark$$

$$= 16.67 \quad = 16.80 \rightarrow \checkmark$$

$$(iii) \text{ Variance} = \text{Var}(X) = np(1-p)$$

$$= 60 \times \frac{5}{18} \left(1 - \frac{5}{18}\right) \text{ OR } 60 \times 0.28 \times (1 - 0.28) \rightarrow \checkmark$$

$$= \frac{325}{21} \quad = 12.10 \rightarrow \checkmark$$

$$= \boxed{12.04} \text{ (2 d.p.)}$$

# QUESTION 14

Marker's Comments

①  $3 \sin A - 4 \cos A - 4$

$$= 3 \times \frac{2t}{1+t^2} - 4 \times \frac{(1-t^2)}{1+t^2} - 4$$

$$= \frac{6t - 4 + 4t^2 - 4 - 4t^2}{1+t^2}$$

$$= \frac{6t-8}{1+t^2}$$

②  $3 \sin A - 4 \cos A = 4$

$$3 \sin A - 4 \cos A - 4 = 0$$

$$\therefore \frac{6t-8}{1+t^2} = 0$$

$$6t - 8 = 0$$

$$6t = 8$$

$$t = \frac{4}{3}$$

$$\tan \frac{A}{2} = \frac{4}{3}$$

$$\frac{A}{2} = 0.9272$$

$$A = 1.85 \text{ (2d.p.)}$$

$$0 \leq A \leq 2\pi$$

$$\therefore 0 \leq \frac{A}{2} \leq \pi \text{ (only 1st Q)}$$

\* NOTE: Need to

Check  $A = \pi$  as  
a solution ✓

$$\begin{aligned} \text{LHS} &= 3 \sin \pi - 4 \cos \pi \\ &= 0 + 4 = 4 \\ &= \text{RHS} \end{aligned}$$

$\therefore A = \pi$  is a solution

1 Mark Correct answer

③  $\sqrt{\frac{1-\sin 2x}{1+\sin 2x}} = \frac{1-\tan x}{1+\tan x}$

$$\text{let } x = \frac{\pi}{8} \therefore 2x = \frac{\pi}{4} \quad (\text{since } \tan \frac{\pi}{4} = 1)$$

$$\sqrt{\frac{1-\sin \frac{\pi}{4}}{1+\sin \frac{\pi}{4}}} = \frac{1-\tan \frac{\pi}{8}}{1+\tan \frac{\pi}{8}} \rightarrow 1 \text{ Mark}$$

$$\sqrt{\frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}}} = \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{8}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{8}} \rightarrow 1 \text{ Mark}$$

$$\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} = \tan\left(\frac{\pi}{4} - \frac{\pi}{8}\right)$$

$$\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}} = \tan \frac{\pi}{8}$$

$$\sqrt{\frac{(\sqrt{2}-1)^2}{2-1}} = \tan \frac{\pi}{8}$$

$$\sqrt{2}-1 = \tan \frac{\pi}{8}$$

as required.

1 Mark for correct simplification of right hand side.

Alternatively!

$$\sqrt{2}-1 = \frac{1-\tan \frac{\pi}{8}}{1+\tan \frac{\pi}{8}}$$

$$\sqrt{2} + \sqrt{2} \tan \frac{\pi}{8} - 1 - \tan \frac{\pi}{8} = 1 - \tan \frac{\pi}{8}$$

$$\sqrt{2} \tan \frac{\pi}{8} = 2 - 2\sqrt{2}$$

$$\tan \frac{\pi}{8} = \frac{2-2\sqrt{2}}{\sqrt{2}}$$

$$= \sqrt{2}-1$$

\* Can square both sides & form a quadratic in  $\tan \frac{\pi}{8}$